



Modelling Financial Markets

Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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Acronyms used in this material (1/2)

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- AMH: Adaptive Markets Hypothesis
- BDS: Brock–Dechert–Scheinkman (test)
- BET: Bucharest Exchange Trading (index)
- BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- BNR: Banca Națională a României (the National Bank of Romania)
- BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAPM: Capital Asset Pricing Model
- CD: Chow–Denning (multiple variance-ratio test)
- CME: Chicago Mercantile Exchange
- DFA: Detrended Fluctuation Analysis
- EMH: Efficient Market Hypothesis
- ETF: Exchange-Traded Fund
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- FWER: Family-Wise Error Rate
- GMM: Generalized Method of Moments
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HK: Hong Kong
- IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- MIT: Massachusetts Institute of Technology
- ML: Machine Learning
- OLS: Ordinary Least Squares
- PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- RW1: Random Walk 1 (i.i.d. increments)

Acronyms used in this material (2/2)

- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ TOM: Turn Of the Month
- ▣ TSMOM: Time-Series MOMentum
- ▣ US: United States
- ▣ VAR: Vector AutoRegression
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Outline

Motivation

The Efficient Market Hypothesis

Testing Weak-Form Efficiency

Long Memory: Hurst, R/S and DFA

Adaptive Markets and Time-Varying Efficiency

Semi-Strong Efficiency: Event Studies

Anomalies

Behavioural Finance and Limits to Arbitrage

Efficiency and Machine Learning

AI for Scientific Discovery

Conclusions

References

Can Anyone Beat the Market?

The question of this chapter

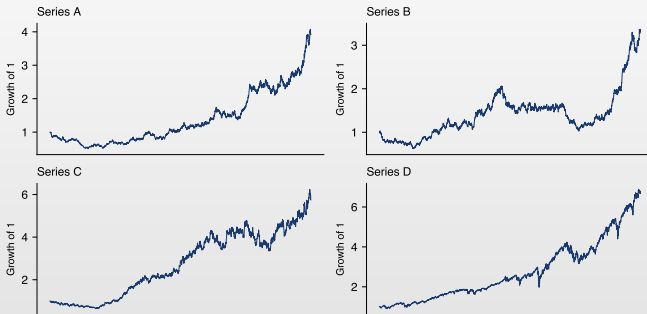
- If past prices could predict future prices, traders would exploit the pattern until it disappeared
- A market in which this has already happened is called **efficient**
- The claim is testable: it restricts the statistical behaviour of returns

What we do in this chapter

- Define efficiency precisely (Fama's three forms)
- Test it: autocorrelation, runs, variance ratios, long memory
- Ask whether efficiency changes over time (adaptive markets)
- Examine anomalies, behavioural finance and machine learning

A Game: Which Series Is the Real Market?

One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



- One panel is the S&P 500 (2010–2026); three are simulated random walks with the same mean and volatility
- Vote before the next slide: A, B, C or D?

The Game: Answer and Lesson

- The real series is **D**; most audiences cannot tell
- **Roberts (1959)** made the same point: random walks produce “patterns” (trends, heads and shoulders) by chance
 - ▶ Chart patterns alone are no evidence of predictability
- A **random walk** is a price whose changes are unpredictable from its own past (defined formally in Section 2)
- We need statistics, not eyes: the rest of the chapter builds them

Pioneers: Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
 - ▶ “The mathematical expectation of the speculator is zero”
- Samuelson (1965): if prices are properly anticipated, their changes are unpredictable
 - ▶ Unpredictability is a sign of a well-functioning market, not of chaos
- Both results are the ancestors of the Efficient Market Hypothesis

Data Used in This Chapter

Group	Series	Period
Developed equity	S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng	2000–2026
Emerging/frontier equity	BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai	2000–2026
Crypto-assets	Bitcoin, Ethereum	2014/2016–2026
Exchange rate	EUR/RON, BNR reference rate	2005–2026

- Daily closing values up to 18 September 2026; returns are daily log returns $r_t = \ln P_t - \ln P_{t-1}$
- Equity indices: weekdays only; days with an unchanged close (holidays filled with the previous price) are removed
- Crypto trades 7 days a week: 365 observations per year
- Each series is analysed on its own calendar; no series is joined to another in this chapter

Key Terms

- ▣ **ACF**: $\hat{\rho}_k = \widehat{\text{corr}}(r_t, r_{t-k})$ as a function of the lag k
- ▣ **i.i.d.**: independent and identically distributed
- ▣ **Heteroskedasticity**: variance that changes over time (volatility clustering, Chapter 1)
- ▣ **bp**: 1 bp = 0.01%
- ▣ **FWER**: the probability of at least one false rejection among many tests
- ▣ **FTSE Russell**: index provider that classifies countries as frontier or emerging

Eugene Fama and the Definition of Efficiency



Eugene F. Fama, Nobel Prize 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

- [Fama \(1970\)](#): a market is **efficient** if prices “fully reflect” all available information
 - ▶ Information set $\mathcal{I}_t$: everything known at time t
- Formally: no strategy using $\mathcal{I}_t$ earns returns above the fair (risk-adjusted) return
- Nobel Prize 2013, shared with Robert Shiller (sceptic, Section 7) and Lars Peter Hansen
- Update and survey: [Fama \(1991\)](#); accessible critique: [Malkiel \(2003\)](#)

Three Forms of Efficiency

Form	Information set $\mathcal{I}_t$	Typical test
Weak	Past prices and returns	Autocorrelation, runs, variance ratios (Sections 3–5)
Semi-strong	All public information (news, reports)	Event studies (Section 6)
Strong	Public and private (insider) information	Returns of insiders and fund managers

- Each form contains the previous one: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak
- This chapter focuses on the weak form; Chapter 3 tests risk-adjusted returns with factor models

Efficiency as a Fair Game: the Martingale

- ▣ A price is a **martingale** if the best forecast of tomorrow's price is today's price
 - ▶ $\mathbb{E}[P_{t+1} | \mathcal{I}_t] = P_t$ (adding a required return μ : $\mathbb{E}[r_{t+1} | \mathcal{I}_t] = \mu$)
- ▣ Equivalently, the excess return $\varepsilon_{t+1} = r_{t+1} - \mu$ is a **fair game**: $\mathbb{E}[\varepsilon_{t+1} | \mathcal{I}_t] = 0$
- ▣ Consequence: $\text{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function g of the past
 - ▶ In particular all autocorrelations $\rho_k = \text{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero
- ▣ The martingale says nothing about the variance: volatility may be predictable (Chapter 1) without contradicting efficiency

Random Walk Hypotheses RW1, RW2, RW3

- Log price $p_t = \ln P_t$ with $p_t = \mu + p_{t-1} + \varepsilon_t$; the three versions differ in what they assume about ε_t [Campbell, Lo & MacKinlay \(1997\)](#)
- **RW1:** ε_t i.i.d. (the strongest; rules out volatility clustering)
 - ▶ Rejected by the stylised facts of Chapter 1
- **RW2:** ε_t independent but not identically distributed (variance may change)
- **RW3:** ε_t uncorrelated: $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$, while $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ is allowed
 - ▶ The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

The Joint-Hypothesis Problem

- ▣ “Fair return” requires a model of expected returns (e.g. CAPM, Chapter 3)
- ▣ Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model [Fama \(1991\)](#)
 - ▶ A rejection may mean an inefficient market **or** a wrong model of risk
- ▣ Example: a strategy earns 5% per year above the index; it may simply carry more risk than the index
- ▣ At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- ▣ Efficiency can never be “proved”, only not rejected by a given test on given data

The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it [Grossman & Stiglitz \(1980\)](#)
 - ▶ Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency
- A testable prediction for our comparison of developed, emerging and crypto markets (Section 3)

Worked Example: How Much Predictability Is “Large”?

- Regress r_t on r_{t-1} : the slope equals $\hat{\rho}_1$ and $R^2 = \hat{\rho}_1^2$
 - ▶ R^2 : share of the variance of r_t explained by the regression
- S&P 500, 2000–2026: $\hat{\rho}_1 = -0.099$, so $R^2 = 0.97\%$
- BET, 2000–2026: $\hat{\rho}_1 = 0.109$, so $R^2 = 1.19\%$
- About 1% of the daily variance is “predictable”: small, but applied every day it can matter
 - ▶ Whether it can be exploited depends on trading costs: statistical and economic efficiency differ

What the EMH Does Not Say

- It does not say prices are “correct” at every moment, only that errors are not predictable
- It does not say nobody beats the market: some will, by luck; the question is whether it persists
- It does not forbid predictable volatility: RW3 allows volatility clustering
- It does not forbid predictable **expected** returns if they compensate time-varying risk
- It does not say bubbles are impossible; it says they are hard to time (Chapter 17)

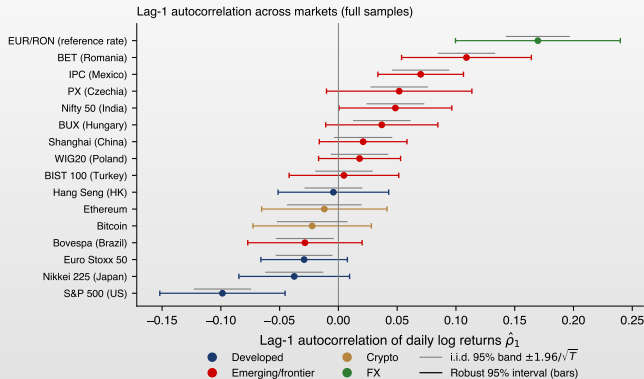
Recap: the Efficient Market Hypothesis

- Efficient market: prices fully reflect the information set $\mathcal{I}_t$ (weak, semi-strong, strong)
- Martingale / fair game: excess returns are unpredictable from $\mathcal{I}_t$
- RW3 is the version to test; RW1 is already rejected
- Every test is joint with a risk model; costly information implies some inefficiency
- Small autocorrelations can be statistically significant and economically irrelevant

Autocorrelation and Its Standard Error

- ▣ Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^T e_t e_{t-k} / \sum_{t=1}^T e_t^2$, with $e_t = r_t - \bar{r}$
- ▣ Under RW1 (i.i.d.): $SE(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
 - ▶ SE: standard error, the standard deviation of an estimator
- ▣ Under RW3 with volatility clustering: $SE(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
 - ▶ $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ (robust variance)
 - ▶ When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow
- ▣ S&P 500: i.i.d. $SE = 0.012$, robust $SE = 0.027$, more than twice as large

Lag-1 Autocorrelation Across Markets



□ Dots: $\hat{\rho}_1$ with robust 95% intervals (bars); grey segments: the naive i.i.d. band

Lag-1 Autocorrelation: Interpretation

- ▣ Positive and significant: BET 0.109 (robust SE 0.028), IPC 0.070, EUR/RON 0.170
 - ▶ Thin trading and slow diffusion of information (BET); a managed exchange rate (EUR/RON)
- ▣ Negative and significant: S&P 500 -0.099
 - ▶ Short-term overreaction and liquidity provision; concentrated in crises (Seminar B7)
- ▣ Bitcoin -0.022 and Ethereum -0.012 : not significant with robust errors
- ▣ Several i.i.d. “rejections” disappear with robust intervals: always use robust inference

Portmanteau Tests: Ljung–Box and Its Robust Version

- ▣ Ljung & Box (1978): $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$ under i.i.d. returns
- ▣ Robust version: $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ under RW3 (Chapter 1)
- ▣ $m = 10$, 5% critical value 18.31:
 - ▶ S&P 500: $Q = 96.3$ but $\tilde{Q} = 19.9$ ($p = 0.031$)
 - ▶ BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
 - ▶ Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- ▣ The classical test rejects too often; the robust test keeps S&P 500 and BET, but not Bitcoin

The Runs Test

- ▣ Nonparametric: uses only the **signs** of returns, so it is immune to heavy tails [Wald & Wolfowitz \(1940\)](#)
- ▣ A **run** is a maximal sequence of equal signs: ++-+++ has 3 runs
- ▣ With n_1 positive and n_2 negative signs, $n = n_1 + n_2$, under independence:
 - ▶ $\mathbb{E}[R] = \frac{2n_1n_2}{n} + 1$, $\text{Var}[R] = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n - 1)}$
 - ▶ $z = (R - \mathbb{E}[R]) / \sqrt{\text{Var}[R]} \approx N(0, 1)$
- ▣ Too **few** runs ($z < 0$): signs persist, positive autocorrelation; too **many** ($z > 0$): reversals

Worked Example: The Runs Test, Step by Step

- ▣ Twelve daily signs: $+ - + - + - + - + - + - +$
- ▣ $n_1 = 7, n_2 = 5, n = 12$; observed runs $R = 11$
- ▣ $\mathbb{E}[R] = 2 \cdot 7 \cdot 5 / 12 + 1 = 6.83$; $\text{Var}[R] = 2.563$
- ▣ $z = (11 - 6.83) / \sqrt{2.563} = 2.60$, two-sided $p = 0.009$
- ▣ $z > 1.96$: reject independence at 5%; too many runs, i.e. signs alternate (reversal)
 - ▶ With $n = 12$ the approximation by the Normal distribution is rough: exact tables of the runs distribution are preferable

Runs Test on Real Markets

- ▣ S&P 500: $z = 4.34$: **more** runs than expected, consistent with $\hat{\rho}_1 < 0$
- ▣ BET: $z = -6.05$: **fewer** runs, signs persist, consistent with $\hat{\rho}_1 > 0$
- ▣ Nifty 50: $z = -5.46$; IPC: $z = -3.27$: persistence in emerging markets
- ▣ Bitcoin $z = 2.68$, Ethereum $z = 3.91$: significant although $\hat{\rho}_1$ is not
 - ▶ Signs and magnitudes carry different information: small reversals, large moves uncorrelated
- ▣ Caveat: the runs test assumes a constant probability of a positive day

The Variance Ratio: Idea

- Under a random walk, variance grows linearly with the horizon: $\text{Var}(r_t + \dots + r_{t-q+1}) = q \text{Var}(r_t)$
 - ▶ q : the horizon in days (e.g. $q = 5$ is one trading week)
- **Variance ratio:** $\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$, equal to 1 under RW1–RW3
- It is a weighted sum of autocorrelations:
 - ▶ $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$
- $\text{VR}(q) > 1$: momentum (positive autocorrelation); $\text{VR}(q) < 1$: mean reversion
- Worked check: S&P 500 $1 + \hat{\rho}_1 = 0.901$ vs estimated $\text{VR}(2) = 0.901$

The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping q -day sums, with bias corrections

$$\blacktriangleright \hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$$

$$\blacktriangleright m = q(T - q + 1)(1 - q/T); \quad \widehat{VR}(q) = \hat{\sigma}_c^2(q) / \hat{\sigma}_a^2$$

- Under RW1 (homoskedastic):

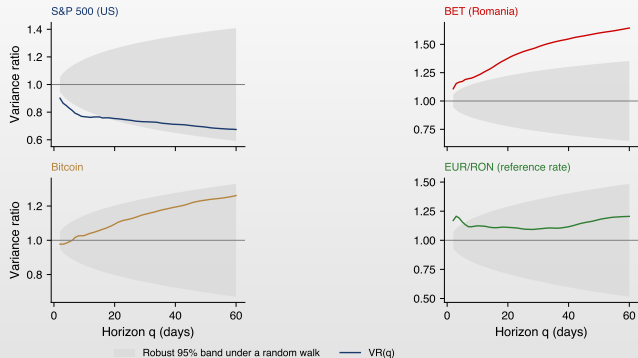
$$\blacktriangleright z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$$

- Homoskedastic: constant variance; this z over-rejects when volatility clusters

The Heteroskedasticity-Robust Statistic z^*

- Under RW3, $\hat{\rho}_k$ has variance $\hat{\delta}_k / T$ with
 - ▶ $\hat{\delta}_k = T \sum_{t=k+1}^T e_t^2 e_{t-k}^2 / \left(\sum_{t=1}^T e_t^2 \right)^2$
- Asymptotic variance of $\widehat{VR}(q)$ and the test statistic:
 - ▶ $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left(\frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \quad z^*(q) = \frac{\sqrt{T} (\widehat{VR}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0, 1)$
- z^* tests RW3: uncorrelated increments with any (bounded) pattern of volatility
- S&P 500, $q = 2$: $\widehat{VR} = 0.901$, $z = -8.12$ but $z^* = -3.64$
 - ▶ Still significant, but the homoskedastic statistic exaggerates the evidence more than twice

Variance-Ratio Profiles



□ $\widehat{VR}(q)$ for $q = 2, \dots, 60$ days; grey: robust 95% band around 1 (from $\hat{\theta}(q)$)

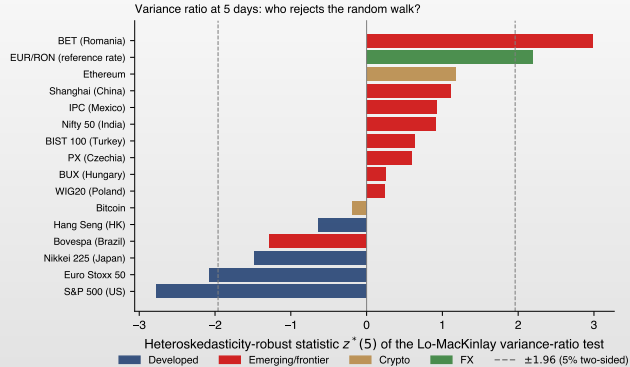
Variance-Ratio Profiles: Interpretation

- ▣ S&P 500: $VR = 0.901, 0.830, 0.753, 0.674$ at $q = 2, 5, 20, 60$: mean reversion at all horizons
- ▣ BET: 1.108, 1.173, 1.377, 1.644: persistence that accumulates with the horizon
- ▣ Bitcoin: 0.977, 0.991, 1.094, 1.261: inside the band up to about one month
- ▣ EUR/RON: 1.170, 1.157, 1.111: short-run persistence of a managed float
- ▣ The band widens with q : long horizons are informative about direction but tests there have low power

Multiple Horizons: Chow–Denning and Choi

- ▣ Testing $q = 2, 5, 10, 20$ separately inflates the probability of a false rejection
 - ▶ Four tests at 5% each: up to $1 - 0.95^4 = 18.5\%$ if independent
- ▣ [Chow & Denning \(1993\)](#): use $CD = \max_i |z^*(q_i)|$ and the studentised maximum modulus distribution
 - ▶ For $m = 4$ horizons, the 5% critical value is 2.49 instead of 1.96
- ▣ [Choi \(1999\)](#): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- ▣ Wild-bootstrap versions for small samples: [Kim \(2006\)](#); applications to Asian markets: [Kim & Shamsuddin \(2008\)](#)

Variance Ratio at One Week, Across Markets



Robust $z^*(5)$ of the Lo-MacKinlay test for all 16 series; dashed lines: ± 1.96

Variance Ratio Across Markets: Interpretation

- Reject at 5% with $z^*(5)$: BET (2.98), EUR/RON (2.19), S&P 500 (-2.77), Euro Stoxx 50 (-2.08)
- Chow–Denning (horizons 2, 5, 10, 20) rejects for 5 of 16 series
 - ▶ BET $p = 0.001$, S&P 500 $p = 0.001$, IPC $p = 0.001$, EUR/RON $p = 0.000$, Ethereum $p = 0.027$
- Bitcoin: $z^*(5) = -0.19$, Chow–Denning $p = 0.835$: no evidence against RW3 over 2014–2026
- No simple ranking “developed = efficient, emerging = inefficient”: the direction differs (reversal vs persistence)

Summary Table: Weak-Form Tests, 2000–2026

Market	T	$\hat{\rho}_1$	robust SE	$\widehat{VR}(5)$	$z^*(5)$	CD p	Runs z
S&P 500 (US)	6 714	−0.099	0.027	0.830	−2.77	0.001	4.34
Euro Stoxx 50	6 836	−0.029	0.019	0.910	−2.08	0.097	3.41
Nikkei 225 (Japan)	6 545	−0.038	0.024	0.923	−1.48	0.317	1.97
Hang Seng (HK)	6 583	−0.004	0.024	0.967	−0.64	0.872	−0.17
BET (Romania)	6 670	0.109	0.028	1.173	2.98	0.001	−6.05
WIG20 (Poland)	6 685	0.018	0.018	1.009	0.24	0.777	2.25
BUX (Hungary)	6 666	0.037	0.024	1.013	0.26	0.427	−0.42
PX (Czechia)	6 682	0.052	0.032	1.041	0.59	0.350	−2.53
BIST 100 (Turkey)	6 692	0.005	0.024	1.033	0.64	0.949	−0.63
Bovespa (Brazil)	6 622	−0.028	0.025	0.928	−1.29	0.521	0.88
IPC (Mexico)	6 689	0.070	0.019	1.039	0.93	0.001	−3.27
Nifty 50 (India)	6 606	0.048	0.024	1.048	0.91	0.178	−5.46
Shanghai (China)	6 474	0.021	0.019	1.046	1.11	0.409	1.02
Bitcoin	4 384	−0.022	0.026	0.991	−0.19	0.835	2.68
Ethereum	3 913	−0.012	0.027	1.067	1.17	0.027	3.91
EUR/RON (reference rate)	5 352	0.170	0.036	1.157	2.19	0.000	−1.66

□ T : number of daily returns; CD: Chow–Denning over $q = 2, 5, 10, 20$; EUR/RON from July 2005, crypto on 365 days per year

Recap: Testing Weak-Form Efficiency

- ▣ Autocorrelations, Ljung–Box, runs and variance ratios all test “no linear predictability from the past”
- ▣ Volatility clustering makes classical standard errors too small: use $\hat{\tau}_k$, $\tilde{Q}$ and z^*
- ▣ Several horizons need a joint test (Chow–Denning); several markets need care in reading “significance”
- ▣ S&P 500 shows reversal, BET and EUR/RON persistence; Bitcoin is not rejected on the full sample
- ▣ Statistical rejection is not yet a profitable strategy: trading costs and risk decide

Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia

[Commons](#)

- Hurst (1951) designed Nile reservoirs: wet years followed wet years more often than chance allows, measured with the rescaled range



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

The Rescaled Range (R/S) Statistic

- For a window of n returns with mean $\bar{r}_n$ and standard deviation s_n :
 - ▶ Cumulative deviations $Y_k = \sum_{i=1}^k (r_i - \bar{r}_n)$, $k = 1, \dots, n$
 - ▶ Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range R_n/s_n
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c n^H$, where H is the **Hurst exponent**
 - ▶ Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging R/S over non-overlapping windows
- $H = 0.5$: no memory; $H > 0.5$: persistence (long memory); $H < 0.5$: anti-persistence
 - ▶ **Long memory**: autocorrelations decay so slowly that their sum diverges, $\rho_k \sim k^{2H-2}$

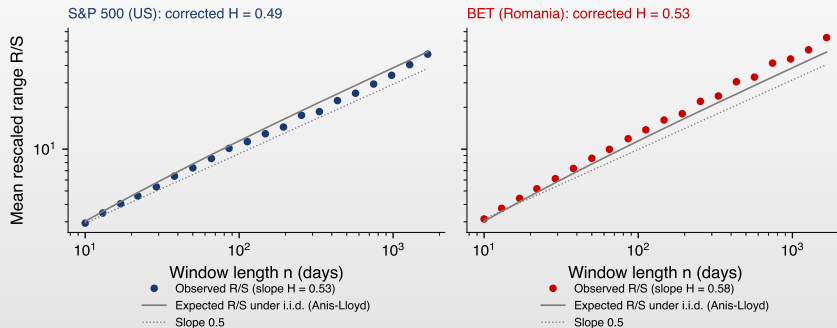
Hurst Exponent and Variance Ratio

- ▣ For self-similar increments (fractional Gaussian noise): $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$
- ▣ Hence $\text{VR}(q) = q^{2H-1}$ and $H = 0.5 + \ln \text{VR}(q) / (2 \ln q)$
- ▣ Worked example, Bitcoin: $\widehat{\text{VR}}(20) = 1.094$
 - ▶ $\hat{H} = 0.5 + 0.0900/5.991 = 0.515$, close to 0.5
- ▣ Same relation for BET ($\widehat{\text{VR}}(20) = 1.377$) is solved in Seminar A5
- ▣ The relation holds only for exact self-similarity: short-memory processes give $H(q)$ that depends on q

The Small-Sample Bias of R/S

- ▣ Even for i.i.d. returns, the slope of $\ln(R/S)$ on $\ln n$ exceeds 0.5 for small n
- ▣ Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. Gaussian returns
 - ▶ $\mathbb{E}[R_n/s_n] = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}$ (Γ : the gamma function)
- ▣ Corrected exponent: $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n] \text{ on } \ln n$
- ▣ Uncorrected estimates overstate memory by about 0.04–0.05 with daily data (next chart)

R/S Analysis: S&P 500 and BET



□ Dots: average R/S for window lengths n ; grey line: the Anis-Lloyd expectation under i.i.d. returns



R/S Analysis: Interpretation

- ▣ S&P 500: raw $\hat{H} = 0.530$, corrected $\hat{H}_{AL} = 0.486$: no long memory, slight anti-persistence
- ▣ BET: raw $\hat{H} = 0.578$, corrected $\hat{H}_{AL} = 0.533$: persistence remains after the correction
- ▣ Without the correction, the S&P 500 would look “persistent” ($0.530 > 0.5$): a pure small-sample artefact
- ▣ Open question: is BET's $H > 0.5$ **long** memory or just the short-run autocorrelation $\hat{\rho}_1 = 0.109$?
 - ▶ Classical R/S cannot tell them apart: next slide

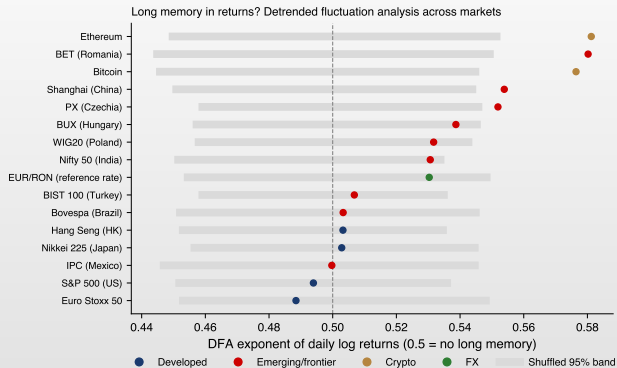
Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Replace s_n by a long-run (HAC) standard deviation with ℓ lags:
 - ▶ $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{\ell} \left(1 - \frac{j}{\ell+1}\right) \hat{\gamma}_j$, $\hat{\gamma}_j$: autocovariance at lag j
 - ▶ $V = \frac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$; bandwidth ℓ chosen as in Andrews (1991)
- Under short memory, V lies in $[0.809, 1.862]$ with probability 95%
- Full samples: S&P 500 $V = 1.379$, Bitcoin 1.530, PX 1.750, EUR/RON 1.429: not rejected
 - ▶ BET $V = 1.961$ ($\ell = 7$) and Ethereum $V = 1.968$: just above 1.862, weak evidence of long memory

Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): robust to slow trends and non-stationary means
- Algorithm:
 - ▶ Profile $y_k = \sum_{i=1}^k (r_i - \bar{r})$; split into boxes of length n
 - ▶ In each box fit a linear trend and keep the residuals
 - ▶ $F(n)$ = root mean square of all residuals; repeat for many n
- $F(n) \propto n^\alpha$: the DFA exponent α plays the role of H ($\alpha = 0.5$: no memory)
- Significance by **shuffling**: random permutations destroy any memory but keep the distribution
 - ▶ The 2.5% and 97.5% quantiles of α over 100 shuffles form a 95% band

DFA Exponent Across Markets



□ Dots: DFA exponent of daily log returns; grey bars: 95% band from shuffled series (no memory)

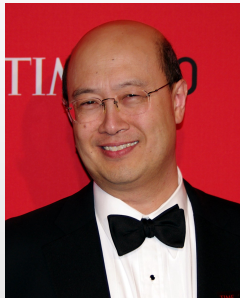
DFA Across Markets: Interpretation

- ▣ Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, Bovespa 0.503, IPC 0.500
- ▣ Above the band: BET 0.580 (band up to 0.551), Ethereum 0.581, Bitcoin 0.576, Shanghai 0.554, PX 0.552
 - ▶ Shuffling destroys **all** dependence, including volatility clustering, which also raises α
- ▣ Cross-check with Lo's V : Bitcoin ($V = 1.530$) and PX ($V = 1.750$) are not rejected, BET and Ethereum marginally are
- ▣ Lesson: long memory in returns is weak and fragile; long memory in volatility (Chapter 1) is strong

Recap: Long Memory

- ▣ The Hurst exponent measures how the range of cumulative returns scales with the horizon: $H = 0.5$ means no memory
- ▣ Classical R/S is biased upward in small windows: always use the Anis–Lloyd correction
- ▣ Short memory mimics long memory: Lo's V separates them
- ▣ DFA with shuffled bands flags BET and crypto, but part of the signal is volatility clustering
- ▣ Developed indices show no long memory in returns

Andrew Lo and the Adaptive Market Hypothesis



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms)
 - ▶ Species adapt by trial and error, not by perfect optimisation
- Efficiency depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time**, and anomalies appear, weaken and may reappear

EMH versus AMH

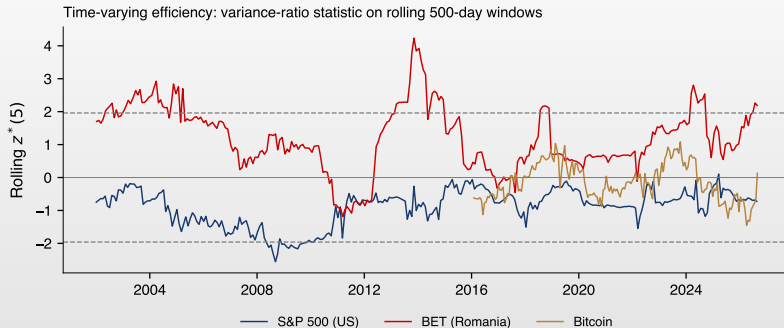
	EMH	AMH
Investors	Rational, optimising	Bounded rationality, heuristics
Efficiency	A state: yes or no	A degree that changes with conditions
Risk–return	Stable trade-off	Changes with the population of investors
Anomalies	Mispriced risk or data mining	Transient profit opportunities
Empirical tool	Full-sample tests	Rolling-window tests

- Evidence of time variation: survey [Lim & Brooks \(2011\)](#); long-run indices [Urquhart & McGroarty \(2016\)](#)

Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
 - ▶ Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- Null band for DFA: simulate 300 i.i.d. Student- t_4 series of 500 observations
 - ▶ 95% band [0.378, 0.614]: much wider than for the full sample
- Caveats
 - ▶ Overlapping windows are strongly correlated: runs of rejections are not independent evidence
 - ▶ Even under the null, about 5% of windows reject: compare the share of rejections with 5%

Rolling Variance Ratio: S&P 500, BET, Bitcoin

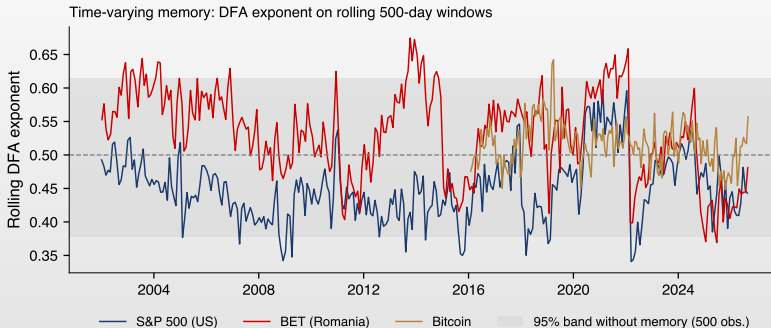


□ $z^*(5)$ on rolling 500-day windows; dashed lines ± 1.96

Rolling Variance Ratio: Interpretation

- ▣ Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- ▣ S&P 500: close to the 5% expected by chance; the full-sample rejection is concentrated in a few turbulent windows
- ▣ BET: rejections cluster in episodes, then fade: the pattern the AMH predicts
 - ▶ Compare the pre- and post-2020 periods in Seminar C1 (FTSE upgrade)
- ▣ Bitcoin, 2014–2026: no 500-day window rejects at 5%

Rolling DFA Exponent: S&P 500, BET, Bitcoin



- DFA exponent on rolling 500-day windows; grey: 95% band without memory

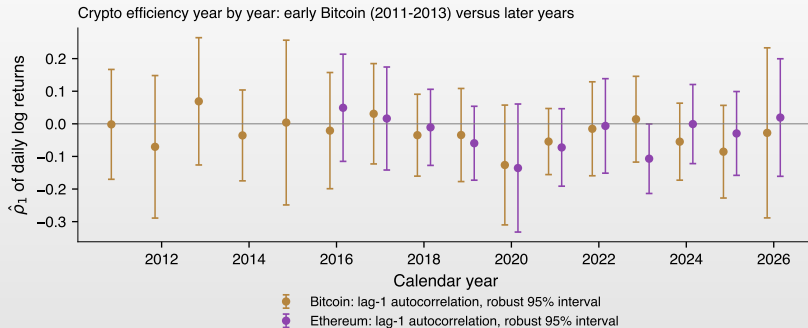
Rolling DFA: Interpretation

- ▣ Share of windows outside $[0.378, 0.614]$: S&P 500 6.4%, BET 12.9%, Bitcoin 1.1%
- ▣ The S&P 500 exponent drifts below 0.5 (anti-persistence) in several windows, matching its negative $\hat{\rho}_1$
- ▣ BET moves in and out of the band: efficiency is a state that changes
- ▣ With 500 observations, the band is wide: rolling DFA detects only strong episodes
 - ▶ Rolling tests trade precision for time resolution

Is Bitcoin Efficient? The Literature

- [Urquhart \(2016\)](#): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
- [Nadarajah & Chu \(2017\)](#): with a power transformation of returns, Bitcoin is weak-form efficient
- [Kristoufek \(2018\)](#): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- [Tran & Leirvik \(2020\)](#): time-varying efficiency across many crypto-assets, increasing over time
- [Pele et al. \(2023\)](#): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets

Crypto Efficiency Year by Year



- Lag-1 autocorrelation of daily returns for each calendar year, with robust 95% intervals; Bitcoin from 2011

Crypto Year by Year: Interpretation

- ▣ Only 1 of the 27 yearly $\hat{\rho}_1$ has $|t| > 1.96$ with robust errors: Ethereum 2023, $\hat{\rho}_1 = -0.107$ (SE 0.054)
 - ▶ Chance alone would produce about $0.05 \times 27 = 1.35$ such years
- ▣ Bitcoin 2011–2013: $\hat{\rho}_1 = -0.002, -0.071, 0.069$; the largest $|z^*(5)|$ in any year is 1.71
- ▣ This differs from [Urquhart \(2016\)](#), who rejects efficiency in early years
 - ▶ Different tests (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S), a different price source and robust vs classical errors
 - ▶ A single year of 365 returns gives a robust SE of about 0.06–0.13: small effects are invisible
- ▣ The evidence against crypto efficiency depends strongly on the test and on the inference

Romania: Frontier, then Emerging

- ▣ FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging, effective 21 September 2020
- ▣ Expected channels: more foreign investors, more analysts, more liquidity $\Rightarrow$ faster price discovery
- ▣ Earlier BVB evidence: crashes and log-periodic bubbles [Pele & Mazurencu-Marinescu \(2012\)](#); herding in an illiquid market [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- ▣ Did efficiency improve? Seminar C1 answers with a before/after test and a block bootstrap
 - ▶ Block bootstrap: resamples blocks of consecutive days to keep the dependence [Künsch \(1989\)](#)

Recap: Adaptive Markets

- ▣ The AMH replaces “efficient or not” with “how efficient, when and why”
- ▣ Rolling tests reveal episodes: BET rejects in 22.4% of windows, the S&P 500 near the 5% chance level
- ▣ Rolling statistics need proper null bands and must be read as correlated evidence
- ▣ Crypto efficiency is test-dependent: our robust yearly tests do not reject
- ▣ Market-structure events (FTSE upgrade) are natural experiments for efficiency

The Event-Study Method

- Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- Market model on an estimation window before the event: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$
 - $r_{m,t}$: market return; b_i : sensitivity of asset i to the market
- Abnormal return $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; cumulative $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- Test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$, with L the length of the event window MacKinlay (1997)
- Efficiency predicts a jump on day 0 and no drift or reversal afterwards

Worked Example: the Romanian Bank Tax of December 2018

- ▣ Seminar 0, Exercise 7: tax announced on the evening of 18 Dec 2018; day 0 = 19 Dec; Banca Transilvania vs BET
- ▣ Market model on days $[-250, -11]$: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$
- ▣ Day 0: $AR_0 = -8.3\%$ ($t = -10.0$); days 1–5 are positive; $CAR[0, +5] = -1.1\%$, $t = -0.54$
- ▣ Most of the day-0 fall reverses within three days: an **overreaction**, against semi-strong efficiency
 - ▶ Or new information (the tax was later softened): the joint-hypothesis problem again

How Fast Is “Fast”?

- Daily data can only show that news is absorbed within a day
- At high frequency, futures prices react to scheduled macro news within microseconds; [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) measure this race at Eurex and CME
- Crypto events: [Pele, Lin, Wang & Gaman \(2025\)](#) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology (Chapter 10, market microstructure)

What Is an Anomaly?

- An **anomaly** is a return pattern that the asset-pricing model cannot explain
- Three families
 - ▶ Calendar: day of the week, January, turn of the month
 - ▶ Time-series: momentum and reversal of an index
 - ▶ Cross-sectional: size, value, momentum across stocks; the “factor zoo” is covered in **Chapter 3**
- Three explanations: mispriced risk (joint hypothesis), behavioural bias, or data mining
- Anomalies weaken once known: portfolio returns are 26% lower out of sample and 58% lower after publication McLean & Pontiff (2016)

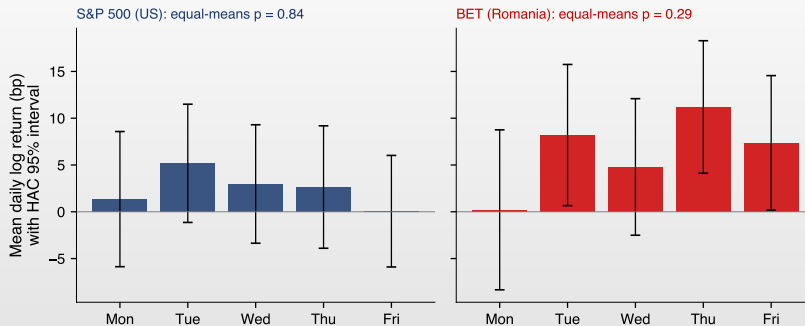
Calendar Anomalies

- **Weekend effect:** negative average returns on Mondays in the US before 1980 French (1980)
- **January effect:** higher returns in January, mostly small stocks Rozeff & Kinney (1976)
- **Turn of the month (TOM):** the last trading day of a month and the first three of the next Ariel (1987); Lakonishok & Smidt (1988)
 - ▶ Proposed channel: salary and pension flows invested at the start of the month
- Test each with a regression on dummy variables (1 on the special days, 0 otherwise)

Regression with HAC Standard Errors

- Day-of-week model without intercept: $r_t = \sum_{d=1}^5 \beta_d D_{d,t} + u_t$
 - ▶ $D_{d,t} = 1$ if day t is weekday d ; β_d is the mean return of weekday d
 - ▶ Wald test of $H_0 : \beta_1 = \dots = \beta_5$ (no weekday effect)
- Residuals u_t are heteroskedastic and autocorrelated: OLS standard errors are wrong
 - ▶ OLS: ordinary least squares
- Newey & West (1987) HAC covariance with L lags (here $L = 5$ days, 3 months for monthly data):
 - ▶ $\hat{S} = \hat{\Gamma}_0 + \sum_{j=1}^L (1 - \frac{j}{L+1})(\hat{\Gamma}_j + \hat{\Gamma}_j^\top)$, $\hat{\Gamma}_j = \frac{1}{T} \sum_t \hat{u}_t \hat{u}_{t-j} x_t x_{t-j}^\top$

Day-of-the-Week Effect: S&P 500 and BET



□ Mean daily log return by weekday (bp), 2000–2026, with HAC 95% intervals

Day of the Week: Interpretation

- ▣ S&P 500: means between 0.1 bp (Friday) and 5.2 bp (Tuesday); equal-means $p = 0.84$
- ▣ No Monday effect since 2000: the anomaly of French (1980) has disappeared
- ▣ BET: Thursday 11.2 bp ($t = 3.10$), Monday 0.2 bp; yet the joint test gives $p = 0.29$
- ▣ A single large t among five days is expected by chance: test the joint hypothesis
 - ▶ Individual t tests against zero also mix the “day effect” with the positive average return

January and Turn-of-the-Month Effects

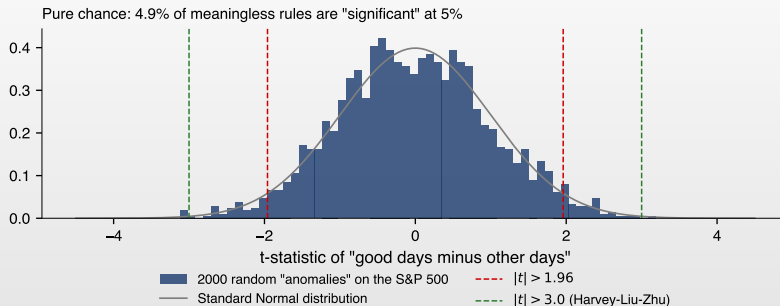
Market	January extra (%/month)	t_{HAC}	TOM extra (bp/day)	t_{HAC}	p
S&P 500 (US)	-0.66	-0.79	3.3	0.95	0.3397
BET (Romania)	3.50	1.49	17.8	3.74	0.0002
Euro Stoxx 50	-0.29	-0.28	5.0	1.18	0.2398
WIG20 (Poland)	0.53	0.40	8.1	1.71	0.0866
BUX (Hungary)	2.66	2.25	14.8	3.58	0.0003

- TOM: BET +17.8 bp per day and BUX +14.8 bp, both with $|t| > 3.5$; S&P 500 only +3.3 bp
- January: large but imprecise for BET ($t = 1.49$); BUX $t = 2.25$ is the only one below $p = 0.05$
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing? Next slides

Multiple Testing: Bonferroni and Holm

- With M independent tests at level α , $\text{FWER} = 1 - (1 - \alpha)^M$
 - ▶ $M = 15$, $\alpha = 5\%$: $\text{FWER} = 53.7\%$, a coin flip
- **Bonferroni**: reject only if $p_i < \alpha/M$; guarantees $\text{FWER} \leq \alpha$ for any dependence
- **Holm**: sort $p_{(1)} \leq \dots \leq p_{(M)}$; reject while $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure
 - ▶ Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month)
- Seminar B6 carries out the full correction

How Many “Anomalies” Does Pure Chance Produce?



- 2000 meaningless rules on the S&P 500: each labels a random 20% of days as “good” and tests good minus other days

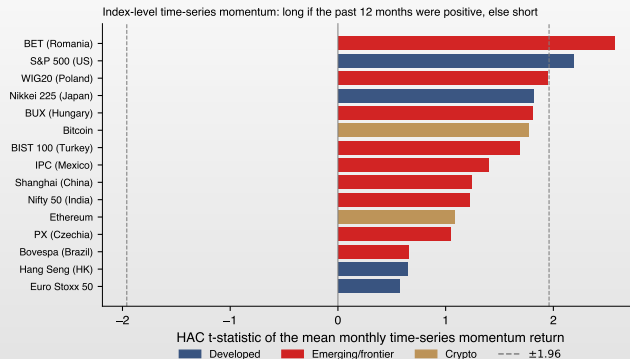
Pure Chance: Interpretation

- 4.9% of the random rules have $|t| > 1.96$, exactly the 5% the test promises; the largest $|t|$ is 3.19
- A researcher who tries many rules and reports the best one will always find a “significant anomaly”
 - ▶ Data snooping in technical trading rules: [Sullivan, Timmermann & White \(1999\)](#)
- [Harvey, Liu & Zhu \(2016\)](#): after hundreds of published factors, a new one should have $|t| > 3.0$; here 0.25% of random rules pass
- BET and BUX turn of the month ($|t| > 3.5$) survive even this stricter threshold

Momentum and Reversal

- **Reversal:** past long-term losers (3–5 years) outperform past winners De Bondt & Thaler (1985)
- **Cross-sectional momentum:** stocks that rose most over 3–12 months keep outperforming Jegadeesh & Titman (1993) (Chapter 3)
- **Time-series momentum (TSMOM)** of a single index Moskowitz, Ooi & Pedersen (2012):
 - ▶ Signal $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$ from the past 12 monthly log returns
 - ▶ Strategy return $s_t r_t$: long after a positive year, short after a negative one
 - ▶ Test: HAC t of the mean of $s_t r_t$ (6 lags); Sharpe ratio $\sqrt{12} \bar{x} / s_x$

Index-Level Time-Series Momentum



□ HAC t -statistic of the mean monthly TSMOM return, 2000–2026 (crypto from 2014/2016)

Time-Series Momentum: Interpretation

- ▣ Significant at 5%: BET ($t = 2.56$, 1.33% per month) and S&P 500 ($t = 2.19$)
- ▣ But the Sharpe ratio barely beats buy-and-hold: BET 0.62 vs 0.61, S&P 500 0.48 vs 0.44
 - ▶ Much of the TSMOM return is the equity premium earned while long
- ▣ BET's monthly autocorrelation is 0.19: the same persistence found in daily data
- ▣ Bitcoin $t = 1.77$ with only 308 months for equities and far fewer for crypto: low power
- ▣ 15 markets tested: the multiple-testing caution applies here too

Recap: Anomalies

- Test calendar effects with regressions on dummies and HAC standard errors; test all days jointly
- The Monday effect is gone; the turn of the month survives in BET and BUX even after Holm correction
- Pure chance produces 5% “anomalies”: correct for the number of tests (Bonferroni, Holm, $|t| > 3$)
- Index momentum is statistically visible in BET and S&P 500 but adds little over buy-and-hold
- Cross-sectional anomalies and the factor zoo: Chapter 3

Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons

- Shiller (1981): stock prices move far more than the dividends that should justify them (excess volatility)
- De Bondt & Thaler (1985): investors overreact; extreme losers become winners over the next 3–5 years

Psychology in Prices

- **Prospect theory** [Kahneman & Tversky \(1979\)](#): outcomes are judged against a reference point and losses loom larger than gains (loss aversion)
 - ▶ Leads to the disposition effect: selling winners too early, holding losers too long
- **Overconfidence**: too much trading and too narrow forecasts
- **Herding**: imitating others, which amplifies booms and crashes [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- **Anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0;

[Wikimedia Commons](#)

- **Shleifer & Vishny (1997)**: real arbitrage is risky and needs capital
- **Noise-trader risk**: mispricing can get worse before it corrects
 - ▶ Noise traders: investors trading on non-information (sentiment, rumours)
- Fund managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Hence anomalies can persist where arbitrage is hard: small, illiquid, frontier markets

Trading Floors and Crowds



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain; [Wikimedia Commons](#)

- Price discovery used to happen among people in one room: information spread by voice
- Today: electronic order books, algorithms and microsecond latencies (Chapter 10)
- Crowds remain: herding and sentiment now travel through social media
- The AMH view: technology changes the “species” and hence the degree of efficiency

Recap: Behavioural Finance

- ▣ Excess volatility and overreaction challenge the EMH; prospect theory and herding offer mechanisms
- ▣ Biases matter for prices only when arbitrage is limited by risk, capital and costs
- ▣ Where arbitrage is hardest (small, illiquid markets like BET) we indeed find the strongest persistence
- ▣ The AMH reconciles both views: rational and biased behaviour coexist and evolve

Can Machine Learning Beat Efficient Markets?

- ML models capture nonlinear patterns that autocorrelation tests miss
 - ▶ ML: machine learning, algorithms that learn a prediction rule from data (Chapter 13)
- [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks give large economic gains over linear regressions in the US cross-section
- Yet their monthly out-of-sample R^2 is below 1%: consistent with an almost efficient market
- Large language models read news and filings: [Tak & Pele \(2025\)](#) forecast stock trends with LLaMA
- AMH reading: every successful model attracts competitors, and its edge decays

ML and Efficiency: Pitfalls

- ▣ **Data snooping**: many models on one dataset produce a “winner” by chance (as in our random-rules experiment)
- ▣ **Look-ahead bias**: using information not available at the time of the forecast
- ▣ **Costs**: high-turnover strategies lose their edge after trading costs
- ▣ **Non-stationarity**: a pattern found in 2000–2015 may be gone by 2026 ([McLean & Pontiff \(2016\)](#))
- ▣ Chapter 13 shows walk-forward validation and the deflated Sharpe ratio to guard against these

An Open Question: Did Spot ETFs Change Bitcoin's Efficiency?

- US spot Bitcoin ETFs trade from 11 January 2024: new arbitrageurs, new liquidity, a new venue for price discovery
- [Kia, Liu, Li, Song & Xu \(2025\)](#): with a fractionally cointegrated VAR on minute data, the ETF led the spot market in price discovery in 2021–2022
 - ▶ after the spot ETFs, the spot market became the dominant venue again
- [Tran & Leirvik \(2020\)](#): crypto efficiency varies over time and has increased: the AMH view of this chapter
- Open: does a change in market structure change weak-form efficiency, and can a before/after design identify it?
 - ▶ confounders in 2024: the April halving, the rate cycle, record inflows

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, DiD, structural break), efficiency measure. Flag any entry you are unsure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match

□ Hypothesis

- ▶ Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in ρ_1 and $VR(q)$, and a variable that separates it from the others.

□ Data and code

- ▶ Read BTC-USD.CC from data/market (close); log returns; ρ_1 with a heteroskedasticity-robust standard error and the Lo-MacKinlay $VR(5)$ with z^* , in two windows of equal length around 11 January 2024.
- ▶ Replicate a known number first: the chapter's full-sample Bitcoin $\hat{\rho}_1 = -0.022$ (robust SE 0.026)

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: window lengths, weekday-only closes, $VR(q)$ for $q = 2$ to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).

▣ Critique

- ▶ Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in ρ_{01} and name a better outcome variable.

▣ Report

- ▶ Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
- ▶ Prompts that shaped a decision go into `AI_USE.md`

- ▣ The assistant drafts the tests; the human decides which one answers the question

What the Human Must Check

- **i.i.d. standard errors:** $1/\sqrt{T}$ rejects too often under volatility clustering
 - ▶ test: robust SE and z^* , as in the weak-form tests of this chapter
- **Unequal windows:** different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
 - ▶ test: the regulatory date only, plus a placebo distribution at dates without an event
- **Confounding:** before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
- **Many statistics:** ρ_1 , VR(2–20), runs; test: Chow–Denning or Holm, not the best single p -value
- **Low power:** report the minimum detectable change, ≈ 2.8 standard errors of the difference

Mini-Case: Did Bitcoin Become More Efficient?

- A typical assistant proposal: “spot ETFs brought arbitrageurs, so Bitcoin became more efficient”
- Test: two windows of 982 days around 11 January 2024; $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$
 - ▶ $z = \Delta \hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, robust standard errors s
- Before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), $\text{VR}(5) = 1.000$ ($z^* = 0.00$)
- After (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.049$), $\text{VR}(5) = 0.939$ ($z^* = -0.64$); $\Delta \hat{\rho}_1 = -0.014$, $z = -0.21$
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest $|z| = 1.57$
- Interpretation: no rejection in either window, so there was no inefficiency for the ETFs to remove
 - ▶ minimum detectable change in ρ_1 : 0.18; the question needs another outcome, such as intraday price discovery

 [ai_discovery_case.py](#)

Your Turn: A Project Seed

- **Question:** did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
 - ▶ rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
- **Replicate first:** the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , dates, windows, corrections) committed before the results
 - ▶ code as a Quantlet; a report with robust inference, placebo tests and power
- **Logs:** prompts in AI_USE.md; every caught error (an i.i.d. standard error, an invented reference) in AI_ERRORS.md

Conclusions

- Efficiency means unpredictable excess returns given the information set; weak-form tests are the cleanest
- Robust inference is essential: volatility clustering makes classical tests reject too often
- 2000–2026: S&P 500 shows mild reversal, BET and EUR/RON persistence, Bitcoin no robust rejection
- Long memory in returns is weak; apparent memory is often short memory or volatility clustering
- Efficiency varies over time (AMH); anomalies must survive multiple-testing corrections and costs
- Statistical rejection $\neq$ exploitable profit $\neq$ irrational market

Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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